Electrostatic–hydrophobic cooperation in the mesoscale assembly of synapsin and synaptic vesicles

Short title: Force control of synapsin–vesicle assemblies

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Abstract

Synapsin associates with synaptic vesicles, but local association does not always connect vesicle groups. We used a three-dimensional Brownian agent model to compare electrostatic and hydrophobic interactions in a four-condition design at fixed abundance and confinement. Electrostatics rapidly placed synapsin near almost every vesicle, while larger networks developed more slowly. With both interactions active, a large connected assembly formed earlier and synapsin motion was slower. Following all conditions for 84 ms showed continued growth under either interaction alone. Their vesicle participation became similar despite different levels of protein association. The positive early interaction contrast decreased as these assemblies grew, and the later contrast depended on which structural measure was used. Contact composition and network connectivity therefore resolve different stages of synapsin-mediated organization, from local vesicle coverage to collective assembly.

Author summary

Synapsin helps organize the vesicles that store neurotransmitters. We asked whether the interactions that bring synapsin close to vesicles also connect vesicles into larger groups. In a computer model, electrical interactions quickly placed synapsin near almost every vesicle. The vesicles took longer to form large connected groups. Adding hydrophobic attraction produced a larger assembly earlier and reduced synapsin motion. When we followed the simulations for longer, both single-interaction conditions continued to grow. They eventually included similar fractions of vesicles in their largest groups, although their protein association remained different. Counting local protein contacts and measuring connections between vesicles describe different aspects of assembly. Their relationship also changes as the system develops.

1 Introduction

A presynaptic terminal holds synaptic vesicles (SVs) in groups while allowing them to move and participate in neurotransmission. The protein-rich SV membrane offers many sites for local interactions [1]. As vesicles and proteins move, these interactions can also link vesicles into larger assemblies.

Experiments on synapsin have linked molecular interactions to vesicle organization. Early bilayer experiments identified both electrostatic association and hydrophobic engagement of the synapsin head domain [2]. Purified-vesicle experiments distinguished lipid binding from an ionic-strength- and phosphorylation-sensitive protein-associated contribution [3]. Computational cage and cytoskeletal cross-linking models then examined how phosphorylation-dependent binding could redistribute vesicles [4]. Later, perturbation of an amphipathic lipid-packing sensor motif linked membrane curvature sensing to synapsin binding and vesicle clustering [5]. Binding and clustering therefore involve more than one type of interaction.

Reconstitution experiments showed that synapsin assemblies capture vesicles and disperse following kinase treatment [6]. Perturbing the disordered region of synapsin also affects vesicle clusters in a living lamprey synapse [7]. Single-molecule tracking linked vesicle sequestration to restricted synapsin motion [8]. The mechanism also depends on the synaptic context. Song and Augustine distinguished synapsin-dependent condensation at inhibitory synapses from tetramer-mediated cross-linking at excitatory synapses [9]. Reserve-pool tracking also implicated Synapsin 2a tetramerization in vesicle organization [10]. These findings motivate separate measures of protein association, vesicle connectivity and particle motion.

Microscopy and a binding-particle model related assembly size and morphology to the synapsin-to-lipid ratio and vesicle adhesion [11]. X-ray scattering and cluster simulations resolved noncompact SV–synapsin assemblies with close intervesicle contacts [12]. More recent X-ray microrheology of synapsin assemblies with colloid-supported lipid bilayers found concentration-dependent motion and relaxation [13]. Together, these studies measure assembly structure and dynamics at different scales.

Simulation approaches span several resolutions. Atomistic synapsin simulations coupled to experiments revealed an ATP-dependent conformational route to vesicle binding and oligomerization [14]. Residue-level models resolve charge and hydropathy along protein chains [15], while coarse-grained simulations of repetitive peptides show that the balance of electrostatic repulsion and hydrophobic attraction depends on sequence and solution conditions [16]. At the particle level, the synapsin–vesicle model of Alfken et al. addresses binding redistribution and finite assembly size [11]. An agent model can represent thousands of moving proteins and vesicles while retaining their identities and pair interactions. We use this resolution to test how electrostatic and hydropho-

bic interactions change collective organization at fixed abundance and confinement. Random-graph and continuum-percolation studies show how connectivity can change nonlinearly with the number and organization of edges [17,18]. This framework lets us ask how local synapsin association relates to the wider vesicle population.

We vary electrostatic and hydrophobic interactions independently in a three-dimensional Brownian agent model. Particle abundance, confinement and dispersion attraction remain fixed. Species-resolved contact networks measure protein association and vesicle connectivity. Saved trajectories also allow a mobility comparison with published tracking data. The comparison shows how the two interactions shape local association and collective vesicle organization.

2 Results

The four conditions differ in the electrostatic (E) and hydrophobic (H) interactions. Particle population and confinement are the same. Each condition has three independent simulation runs, followed for 84 ms. Structural comparisons use the final 180 saved frames, from 58.94 to 84 ms. We also retain the earlier 28-ms comparison to examine how the response develops with time.

2.1 Combined interactions produce an earlier and larger connected assembly

With neither E nor H active, the late mean all-agent largest-component fraction was 0.0368. It reached 0.3440 with H alone, 0.5025 with E alone and 0.9200 with both interactions (table 1 and Fig 1). The combined-interaction condition had the largest component in every seed at both observation horizons.

Table 1: Structural outcomes over 58.94–84 ms at the primary 2-nm association shell.

Condition	All-agent largest component	Non-isolated Syn	SVs in largest component
E0H0	0.0368 (0.0150)	0.3172 (0.0035)	0.1143 (0.0492)
E0H1	0.3440 (0.0667)	0.5678 (0.0221)	0.5229 (0.0783)
E1H0	0.5025 (0.0847)	0.7639 (0.0142)	0.5323 (0.0728)
E1H1	0.9200 (0.0064)	0.9516 (0.0023)	0.8909 (0.0204)

Entries are means across three independent seeds with between-seed SD in parentheses. E and H denote the electrostatic and hydrophobic kernels; van der Waals attraction is present in every row.

The time courses show a clear difference in how quickly the assemblies developed (Fig 2A). At 28 ms, the mean all-agent largest-component fractions were 0.020, 0.090, 0.099 and 0.610 for E0H0,

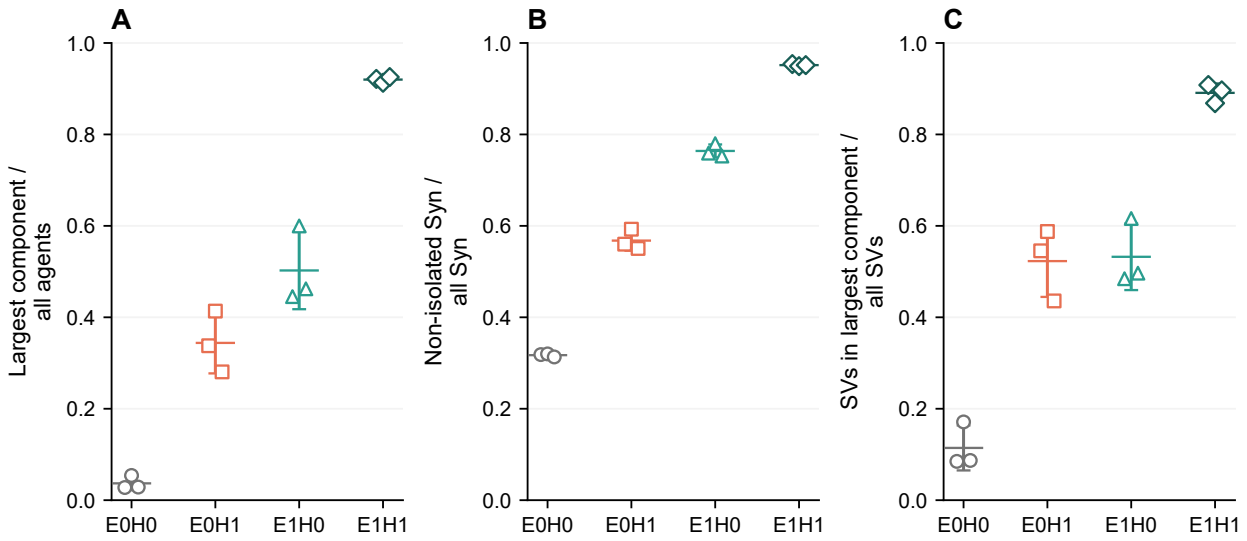


Fig 1: Late-window responses to the electrostatic (E) and hydrophobic (H) interactions at the 2-nm graph shell, averaged over 58.94–84 ms. Open symbols show individual runs; horizontal marks and bars show the mean and one between-seed SD ($n = 3$ per condition). Non-isolated synapsin includes Syn–Syn and Syn–SV components. Van der Waals attraction is active throughout.

E0H1, E1H0 and E1H1. At 84 ms, they were 0.034, 0.409, 0.595 and 0.926. The E1H1 growth slowed at later times: its mean fitted slope decreased from 0.0148 per ms over 14–28 ms to 0.00038 per ms over 70–84 ms (Fig 2C). The single-interaction conditions continued to grow during the extended observation.

For each seed, we compared joint activation with the sum of the separate E and H effects, using E0H0 as the common background:

$$\Delta_{EH} = (Y_{11} - Y_{10}) - (Y_{01} - Y_{00}). \tag{1}$$

For the all-agent largest-component fraction, the early late-window contrast over 19.74–28 ms was 0.3723 and positive in all three seeds. Over 58.94–84 ms, its mean decreased to 0.1103, with seed values 0.1660, −0.0463 and 0.2112. Growth under E or H alone reduced the initial separation from their additive reference.

2.2 Protein association and vesicle participation follow different time courses

Electrostatics produced the larger increase in the non-isolated synapsin fraction Q_{Syn}. At 84 ms, its late-window mean was 0.7639 with E alone and 0.5678 with H alone (table 1). The corresponding

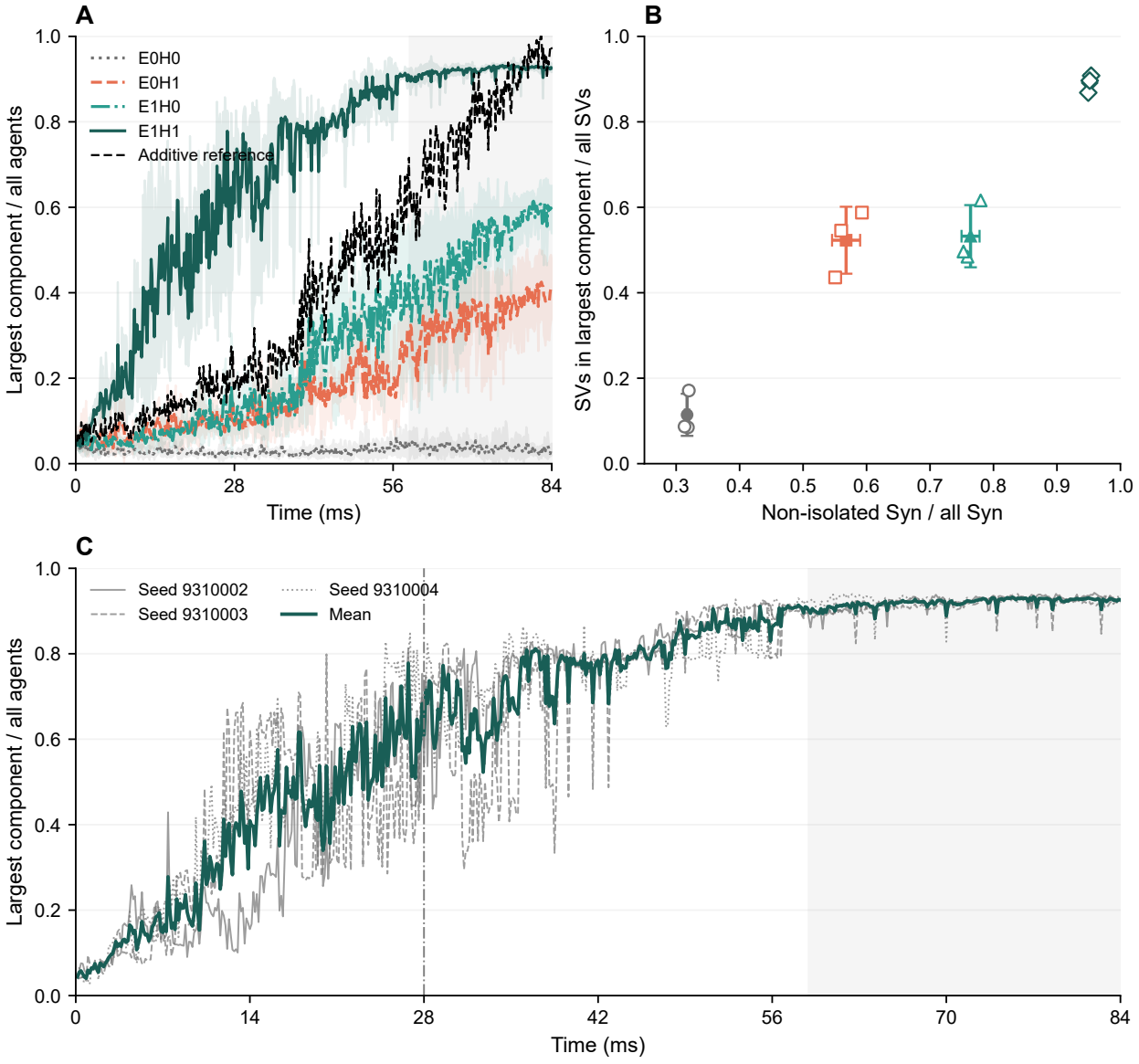


Fig 2: Network growth and vesicle participation at the 2-nm graph shell. (A) All-agent largest-component fraction for all four conditions through 84 ms. Lines show condition means, bands show one between-seed SD, and the dashed line is the additive reference $\overline{L}_{10}(t)+\overline{L}_{01}(t)-\overline{L}_{00}(t)$. Shading marks the final 180 frames. (B) Late-window non-isolated synapsin versus SV participation in the largest component. Open symbols show runs; filled symbols and bars show means and one SD. (C) The three E1H1 trajectories and their mean. The vertical line marks 28 ms; shading marks 58.94–84 ms.

SV fractions in the largest component, L_{SV}, were similar: 0.5323 and 0.5229. The two conditions therefore had different protein participation despite comparable collective vesicle participation.

This relationship developed over time. In the earlier 19.74–28-ms window, E alone gave more non-isolated synapsin than H alone (0.6989 versus 0.5157), but fewer SVs in the largest component (0.1273 versus 0.1839). That ordering of vesicle participation held in every seed at 28 ms. The longer observation shows that it describes an early stage of assembly.

With both interactions active, the 84-ms late-window means reached 0.9516 for Q_{Syn} and 0.8909 for L_{SV}. Their E-by-H contrasts were -0.0628 and -0.0500, respectively. For L_{SV}, the seed contrasts were -0.0369, -0.1642 and 0.0509, compared with three positive values and a mean of 0.3811 at 28 ms. The conditional differences are given in table 2. The interaction contrast thus depends on the structural observable and the observation period.

Table 2: Paired seed-block contrasts over 58.94–84 ms at the 2-nm shell.

Outcome	Conditional contrast	Mean difference (SD)
All-agent largest component	H effect at E0	0.3072 (0.0538)
	H effect at E1	0.4175 (0.0906)
	E effect at H0	0.4656 (0.0698)
	E effect at H1	0.5760 (0.0730)
Non-isolated Syn	H effect at E0	0.2506 (0.0194)
	H effect at E1	0.1877 (0.0159)
	E effect at H0	0.4466 (0.0115)
	E effect at H1	0.3838 (0.0238)
SVs in largest component	H effect at E0	0.4086 (0.0562)
	H effect at E1	0.3586 (0.0929)
	E effect at H0	0.4180 (0.0239)
	E effect at H1	0.3680 (0.0900)

Values are mean differences in fractions, with the SD of the three paired seed differences in parentheses. "H effect" means H1–H0 at fixed E; "E effect" means E1–E0 at fixed H.

2.3 Local vesicle coverage and connections between groups

We separated Syn–Syn, Syn–SV and SV–SV contacts in the saved frames. At the 2-nm shell, the 84-ms late-window fraction of synapsin next to at least one SV was 0.1906 in E0H0, 0.4647 in E0H1, 0.7508 in E1H0 and 0.9499 in E1H1 (Fig 3).

Electrostatics alone already gave broad SV coverage at 28 ms: 99.72% of SVs had a synapsin neighbor, although only 12.73% belonged to the largest component. At 84 ms, the corresponding late-window values were 99.14% and 53.23%. Extensive local coverage therefore preceded collective

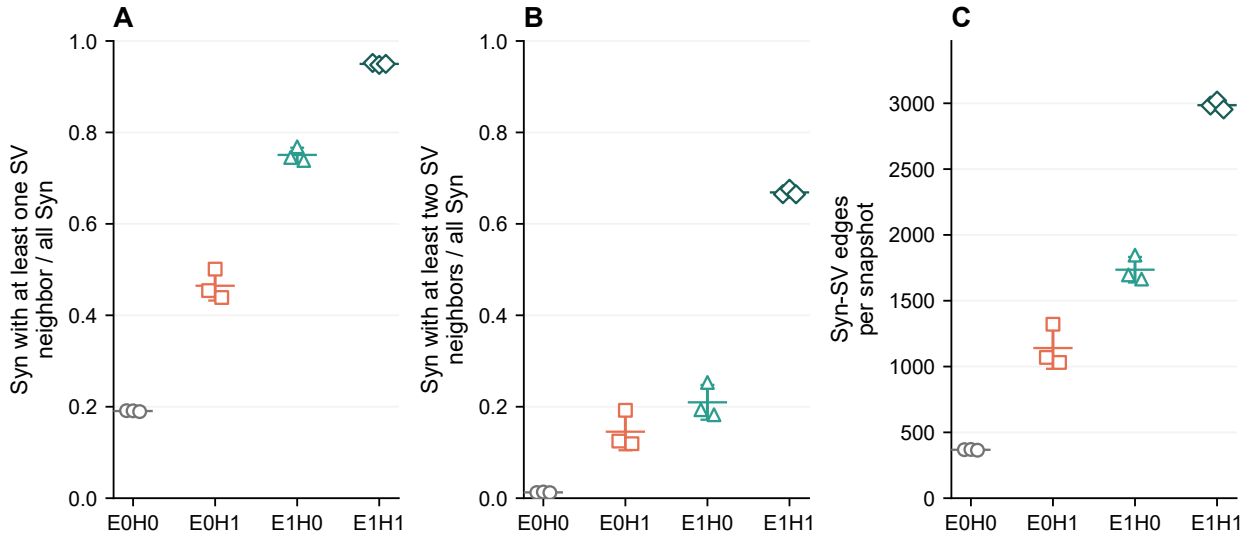


Fig 3: Species-resolved contacts over 58.94–84 ms at the 2-nm graph shell. (A) Fraction of Syn agents adjacent to at least one SV. (B) Fraction adjacent to at least two distinct SVs. (C) Number of Syn–SV edges per snapshot. Open symbols show individual late-window run means; marks and bars show the mean and one between-seed SD ($n = 3$ per condition). Contact uses outer radii, including the SV corona.

vesicle connectivity. In E1H1, 98.08% of SVs had a synapsin neighbor and 89.09% belonged to the largest component in the later window.

The late-window fraction of synapsin next to at least two SVs was 0.1454 with H alone, 0.2097 with E alone and 0.6688 with both interactions (Fig 3B). Mean Syn–SV edge counts per snapshot were 1,140.6, 1,735.7 and 2,985.4, respectively. The joint condition also had more Syn–Syn edges, with a mean of 1,372.8, compared with 469.2 under H alone and 314.5 under E alone. Its many multivesicle neighbors provide geometric connections between vesicles within the association graph.

The single-interaction conditions differed in contact composition. E1H0 had more total edges than E0H1 (2,082.0 versus 1,793.9 per snapshot), but fewer Syn–Syn and SV–SV edges. Their similar L_{SV} values at 84 ms therefore arose from different local contact networks. Their all-agent largest-component fractions remained more distinct because that measure also counts synapsin particles.

2.4 The joint condition retains the largest assembly across observation shells

Across 0, 2, 4 and 6 nm surface-gap shells, E1H1 retained the largest mean all-agent component and largest-component SV fraction at 84 ms. Its all-agent fraction ranged from 0.8872 to 0.9382,

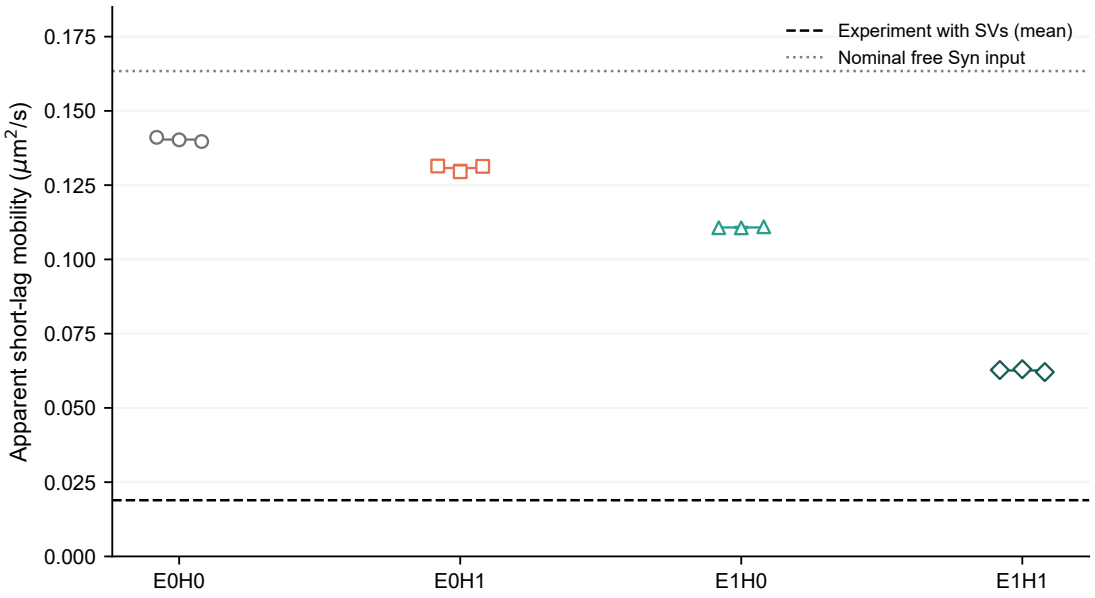


Fig 4: Short-lag synapsin mobility over the full 84-ms trajectories and the experimental reference. Symbols show run means over synapsin agents; marks and bars show the between-seed mean and one SD. The apparent coefficient is one quarter of the free-intercept slope of projected time-averaged MSD over 2.10–3.92 ms. The dashed line is the Hoffmann et al. with-SV mean from released Fig. 2h source data [8]. The dotted line shows the nominal unconfined synapsin diffusion coefficient.

compared with 0.0185–0.1433 for E0H0. The ordering was consistent across these graph thresholds applied to the same trajectories.

2.5 Stronger assembly accompanies slower short-lag motion

We estimated synapsin mobility from saved positions projected onto the xy plane. Apparent diffusion coefficients came from fits to each agent's time-averaged mean-squared displacement (MSD) over lags of 2.10–3.92 ms. For the full 84-ms trajectories, across-seed means of the agent-mean apparent coefficient were 0.1403, 0.1308, 0.1108 and 0.0626 $\mu m^2/s$ for E0H0, E0H1, E1H0 and E1H1, respectively (Fig 4). Mobility was about 55% lower in E1H1 than in E0H0.

The released Hoffmann et al. source data provide an experimental scale reference. They report a mean short-time synapsin coefficient of 0.0189 $\mu m^2/s$ with SVs, versus 0.0253 $\mu m^2/s$ without SVs [8]. The E1H1 full-trajectory mean is about 3.3 times the experimental with-SV mean. Restricting the simulation analysis to 58.94–84 ms gives an E1H1 mean of 0.0548 $\mu m^2/s$. Both the experimental and simulated comparisons show lower synapsin mobility in the more assembled setting. S1 Text gives the per-run estimates and their window sensitivity.

3 Discussion

3.1 Local association and collective organization develop on different time scales

The extended trajectories separate early vesicle coverage from the subsequent growth of connected assemblies. Electrostatics alone placed synapsin near almost every vesicle by 28 ms. At that stage, most vesicles still belonged to separate groups. Their participation in the largest component increased over the next 56 ms while local coverage remained high. With both interactions active, a large connected assembly formed earlier and its growth slowed towards 84 ms.

The pair forces help explain the contact networks (table 3). Electrostatics attracts Syn–SV pairs and repels like-species pairs. Hydrophobic attraction acts on Syn–Syn and Syn–SV pairs. E1H0 favored Syn–SV contacts, while E0H1 had more Syn–Syn and SV–SV contacts. At 84 ms, the two conditions had similar fractions of vesicles in the largest component despite different levels of synapsin participation. Joint activation produced many more synapsin particles close to several vesicles, which can connect vesicles in the association graph.

Table 3: Pair-level action of the two interaction terms at positive outer-surface gap.

Pair class	Electrostatic interaction when E1	Hydrophobic interaction when H1
Syn–Syn	Repulsive	Attractive
Syn–SV	Attractive	Attractive
SV–SV	Repulsive	Zero amplitude

Signs describe radial forces for the assigned effective surface potentials. Dispersion attraction acts in all three pair classes and remains enabled in every condition.

Connectivity changes nonlinearly with the number and placement of edges, as in random graphs and continuum percolation [17,18]. Here, the large positive interaction contrast at 28 ms decreased as the single-interaction assemblies grew. At 84 ms, the all-agent contrast was positive on average with mixed seed signs, while the mean vesicle-participation contrast was slightly negative. The joint condition still had the largest assembly. These results connect the early cooperative response to the timing of network growth and to the structural measure used. Experiments that follow local vesicle coverage and group connectivity over time could distinguish these stages during perturbations of protein–membrane and protein–protein interactions.

3.2 Relation to molecular and particle-level models

The particle model of Alfken et al. links redistribution of vesicle-bound synapsin to adhesion and finite assembly size [11]. Their two-dimensional periodic Langevin model follows proteins on vesicle surfaces. Our model follows freely moving effective synapsin particles inside a three-dimensional compartment and separates electrostatic from hydrophobic interactions. The two models address different routes from local interaction to vesicle assembly.

Residue-level simulations describe charge and hydropathy along protein chains [15]. In repetitive peptides, hydrophobic attraction competes with same-sign electrostatic repulsion [16]. Our two-species model represents this competition for Syn–Syn pairs and includes attraction between oppositely charged Syn–SV pairs. The latter favors local protein association, while larger vesicle assemblies also depend on how the contacts connect separate groups.

Atomistic simulations and experiments connect an ATP-dependent synapsin conformational transition to vesicle binding and oligomerization [14]. At the mesoscale, our contact-network analysis shows how the balance of effective interactions changes the organization of many vesicles.

3.3 Experimental context and biological implications

Reconstitution and neuronal experiments establish synapsin as a regulator of vesicle organization [6, 7]. Scattering measurements show noncompact synapsin–vesicle clusters [12]. Their structure makes connected-component measures relevant alongside cluster size. Studies of synaptophysin tails also support electrostatic interactions in protein–vesicle assembly [19, 20].

Joint activation reduced short-lag synapsin mobility by about 55% relative to E0H0. Hoffmann et al. likewise observed restricted synapsin motion in the presence of vesicles [8]. The simulated apparent diffusion coefficient was about 3.3 times their with-SV mean. A quantitative mobility match would require further calibration.

The distinction between coverage and connectivity may also help interpret synapsin dysfunction. Human *SYN1* loss-of-function variants associated with autism and epilepsy impair vesicle-pool size and trafficking [21]. Oxidation and persulfidation have been linked to different changes in synapsin condensation and neurotransmission [22]. Together, these studies motivate experiments that measure local vesicle association and network connectivity.

3.4 Scope and limitations

All four conditions were followed for 84 ms. Growth of the E1H1 largest component slowed markedly, while the single-interaction conditions were still developing. The reported contrasts describe assembly over a common observation period. Their change between 28 and 84 ms makes observation time part of the comparison. Three seed blocks give limited precision for the later interaction contrasts. We studied one concentration and compartment volume; other concentrations may change the response near a connectivity threshold or saturation.

The agents are spheres with effective surface interactions in a closed compartment. Switching E or H changes several pair classes at once. Pair-specific perturbations would resolve their separate contributions. Graph edges describe proximity. A synapsin near several vesicles is counted as a multivesicle contact; molecular binding is unresolved. The model omits explicit valency, phosphorylation, membrane deformation, active vesicle cycling and hydrodynamic coupling. Numerical checks establish geometric admissibility; convergence of the reported observables at smaller integration steps remains to be assessed. Connectivity and short-lag motion characterize assembly here. A claim of liquid–liquid phase separation would also need evidence for phase coexistence, exchange and system-size behavior.

The mobility reference has different sampling: Hoffmann et al. selected condensate-core tracks, while our estimate includes every synapsin agent and uses a different saved cadence without localization noise. The 3.3-fold difference in apparent mobility is a scale comparison, pending calibration against matched tracking conditions. The interaction switches likewise do not represent specific disease variants or biochemical modifications.

Conclusion

Electrostatic and hydrophobic interactions shape both the timing and the structure of synapsin–vesicle assembly. Electrostatics rapidly gave broad local vesicle coverage. Collective connectivity developed more slowly under either interaction alone, while their combination produced a larger assembly earlier and lower short-lag synapsin mobility. Following all conditions to 84 ms resolved how the early interaction contrast changed as the separate assemblies grew. Local contact abundance, particle composition and vesicle connectivity describe complementary aspects of this process.

4 Methods

4.1 Simulation design and numerical settings

The four force conditions each have three independent seed blocks (9310002–9310004), giving 12 trajectories. Every run contains 200 SVs and 1,807 synapsin agents in a closed 0.1 μm^3 compartment. Each trajectory comprises 120,000 integration steps and 600 saved frames. The original 28-ms trajectories were continued from checkpoints after inspection showed continuing growth. Physical parameters and the random number generator state were retained; duplicate join frames were counted once. The numerical settings are common to all conditions. S1 Text gives the per-run numerical diagnostics.

4.2 Compartment, size distributions, and population

We represent SVs and synapsin as spherical agents in three dimensions, following the scale of synapsin–vesicle reconstitution and neuronal imaging [6, 8]. The closed spherical compartment has volume $V = 0.10\,\mu\text{m}^3$, corresponding to radius

$$R_{\text{b}} = \left(\frac{3V}{4\pi}\right)^{1/3} = 287.94\,\text{nm}. \tag{2}$$

The scale is informed by three-dimensional hippocampal bouton reconstructions [23]. An SV has membrane-core radius b_i, corona thickness $s_i = 4.8\,\text{nm}$, and outer radius $a_i = b_i + s_i$. Synapsin has $b_i = a_i = 3.972\,\text{nm}$ and $s_i = 0$. The protein corona is interpenetrable [12]; the synapsin radius sets the effective particle size.

SV core radii are independently rejection-sampled from a normal distribution with location $\mu_R = 20.45\,\text{nm}$ and input SD $\sigma_R = 2.5\,\text{nm}$, restricted to $15 \le b_i \le 30\,\text{nm}$. The mean and SD inputs follow the excitatory-SV diameter measurements of Tao et al. [24]; the bounds are informed by cryo-electron microscopy (cryo-EM) and small-angle X-ray scattering (SAXS) size measurements [25]. These values specify the distribution before truncation. Synapsin counts are obtained from

$$N_{\text{Syn}} = \left\lfloor N_{\text{A}} C_{\text{Syn}} V + \frac{1}{2} \right\rfloor. \tag{3}$$

Here N_{A} is Avogadro's constant, with C_{Syn} in mol/m^3 and V in m^3; $\lfloor x+1/2 \rfloor$ rounds a nonnegative value to the nearest integer. The chosen 30 µmol/L concentration gives $N_{\mathrm{Syn}} = 1807$. There are 200 SVs at the canonical volume. Initial centers are sampled uniformly within active cells and restricted to the particle-accessible sphere. Before the physical trajectory, an overlap-relaxation stage reduces initial core intersections. Its stopping criterion is an overlap below 0.05 nm for three checks, or no overlapping pairs; the stage has a 200-iteration ceiling.

4.3 Linked cells and boundary treatment

The supremum outer SV diameter is $d_{\max} = 2(30 + 4.8) = 69.6\,\mathrm{nm}$. For each bounding-box dimension L_d, the grid uses

$$n_d = \max\{1, \lfloor L_d/d_{\max} \rfloor\}, \qquad h_d = L_d/n_d. \tag{4}$$

The canonical sphere has eight cells per bounding-box axis, with $h_d = 71.9853\,\mathrm{nm}$. Cells are assigned by particle center. The force calculation searches the home cell and two neighboring cell layers. Every candidate pair enters the force calculation; no further radial cutoff is imposed.

The largest center distance at a 6-nm outer-surface gap is $d_{\max} + 6 = 75.6\,\mathrm{nm}$, within the two-layer search distance of 143.9706 nm. A particle center is confined to radius $R_{\mathrm{b}} - a_i$ from the compartment center. Out-of-domain proposals are mapped inward by radial folding; the same boundary map is applied after overlap corrections. The resulting boundary approximates a closed reflecting wall.

4.4 Pair interactions

Let $r_{ij} = \|\boldsymbol{x}_i - \boldsymbol{x}_j\|$, $\boldsymbol{n}_{ij} = (\boldsymbol{x}_i - \boldsymbol{x}_j)/r_{ij}$, and $g_{ij} = r_{ij} - a_i - a_j$. A positive radial force is repulsive. The production force definition is

$$\boldsymbol{F}_i = \sum_{j \in \mathcal{N}_i} \mathbf{1}\{g_{ij} \geq 0\} \left(E f_{ij}^{E} + H f_{ij}^{H} + f_{ij}^{W} \right) \boldsymbol{n}_{ij}. \tag{5}$$

Here $\mathcal{N}_i$ is the linked-cell candidate set and $E, H \in \{0, 1\}$ switch the electrostatic and hydrophobic terms. Van der Waals attraction remains active in every condition. At negative outer gap, the

surface-force terms vanish and geometric correction controls core overlap.

The electrostatic kernel is a reduced-potential, screened sphere interaction:

$$f_{ij}^{E} = k_{\mathrm{B}}T\frac{a_i a_j}{\ell_{\mathrm{B}}}\,\varphi_i\varphi_j e^{-\kappa g_{ij}}\left(\frac{1}{r_{ij}^{2}}+\frac{\kappa}{r_{ij}}\right), \tag{6}$$

$$\varphi_i = \frac{e\psi_i}{k_{\mathrm{B}}T}, \quad \ell_{\mathrm{B}} = \frac{e^2}{4\pi\epsilon_0\epsilon_r k_{\mathrm{B}}T}, \quad \kappa^2 = \frac{2e^2 N_{\mathrm{A}}(1000I)}{\epsilon_0\epsilon_r k_{\mathrm{B}}T}. \tag{7}$$

The last expression uses SI quantities and I in mol/L; lengths are then converted to nanometers. At $T = 310.15\,\mathrm{K}$, $\epsilon_r = 74.2$, and $I = 0.15\,\mathrm{mol/L}$, $\kappa^{-1} = 0.77886\,\mathrm{nm}$. SV and synapsin effective potentials are respectively -22.5 and $+16\,\mathrm{mV}$. Negative isolated-SV zeta potentials motivate the former [26]; the latter represents an effective positive protein charge scale. The kernel approximates screened electrostatics in a linearized-Poisson–Boltzmann/Yukawa form.

Hydrophobic attraction is

$$f_{ij}^{H} = -\frac{\epsilon_{ij}k_{\mathrm{B}}T}{\lambda_H}e^{-g_{ij}/\lambda_H}, \qquad \lambda_H = 0.5\,\mathrm{nm}. \tag{8}$$

This follows from $U_{ij}^{H} = -\epsilon_{ij}k_{\mathrm{B}}Te^{-g_{ij}/\lambda_H}$ outside contact. An exponential kernel has a surface-force precedent [27]; its length and pair amplitudes specify the effective short-range attraction. The SV–SV hydrophobic amplitude is zero.

For the fixed dispersion background, the membrane/core radii are used:

$$f_{ij}^{W} = -\frac{A\,b_i b_j}{6(b_i+b_j)\widehat{h}_{ij}^{2}}, \qquad \widehat{h}_{ij} = \max\{r_{ij}-b_i-b_j, h_0\}, \tag{9}$$

where $A = 5\times10^{-21}\,\mathrm{J}$ and $h_0 = 0.3\,\mathrm{nm}$. This non-retarded Hamaker–Derjaguin approximation is informed by dispersion theory [28]; it uses a common effective aqueous background constant. The dispersion surfaces are defined by core radii, with lower-gap regularization to bound the attraction near contact.

4.5 Mobility and discrete stochastic dynamics

Independent-particle Stokes drag gives

$$\gamma_i = 6\pi\eta_{\mathrm{eff}} a_i, \qquad M_i = \gamma_i^{-1}, \qquad D_i = k_{\mathrm{B}} T M_i. \tag{10}$$

Brownian particle updates use diagonal mobility [29]. The effective friction input is $\eta_{\mathrm{eff}} = 0.35\,\mathrm{Pa\,s}$, selected with reference to published mobility scales [8, 30]. The nominal unconfined diffusion coefficients are $0.1634\,\mu\mathrm{m}^2/\mathrm{s}$ for synapsin and $0.02571\,\mu\mathrm{m}^2/\mathrm{s}$ for an SV with $a_i = 25.25\,\mathrm{nm}$.

At each Euler–Maruyama step,

$$\boldsymbol{d}_i^n = M_i \boldsymbol{F}_i(\boldsymbol{x}^n)\Delta t, \qquad c_i^n = \min\left\{1, \frac{a_i/4}{\|\boldsymbol{d}_i^n\|}\right\}, \tag{11}$$

$$\boldsymbol{y}_i = \mathcal{B}_i\left(\boldsymbol{x}_i^n + c_i^n \boldsymbol{d}_i^n + \sqrt{2D_i\Delta t}\,\boldsymbol{\xi}_i^n\right), \qquad \boldsymbol{\xi}_i^n \sim N(\boldsymbol{0}, \boldsymbol{I}_3). \tag{12}$$

For zero drift, $c_i^n = 1$. The limiter acts on deterministic displacement, while the Brownian increment remains Gaussian. $\mathcal{B}_i$ is the boundary map. The next state is the result of mobility-weighted iterative overlap correction applied to $\boldsymbol{y}$. Pair admissibility is

$$r_{ij} \geq b_i + b_j - \delta_0, \qquad \delta_0 = 0.5\,\mathrm{nm}. \tag{13}$$

Thus permitted outer overlaps are 0.5 nm for Syn–Syn, 5.3 nm for Syn–SV, and 10.1 nm for SV–SV. Corrections are accumulated simultaneously, averaged over each particle's violating contacts, reflected at the wall, and repeated with updated cells. The residual tolerance is 0.005 nm, with early stopping and a 128-iteration safety ceiling.

The fixed integration step is $\Delta t = 0.0007\,\mathrm{ms}$. Each trajectory has 120,000 physical moves over 84 ms; snapshots are scheduled every 200 steps, or 0.14 ms. Forces are evaluated at the stored state before movement, aligning saved forces and positions.

4.6 Structural graphs and observables

A nonperiodic undirected graph is constructed independently at each stored frame using $A_{ij}^{(\ell)} = \mathbf{1}\{g_{ij} \leq \ell\}$, $i \neq j$. The primary outer-surface shell is $\ell = 2\,\mathrm{nm}$, with $0, 2, 4, 6\,\mathrm{nm}$ sensitivity

shells. Edges represent geometric association. Let $\mathcal{C}_f$ be the connected components at frame f, $n_c = |c|$, and c_f^* the component with the largest total agent count. The species-resolved structural observables are

$$L_{\mathrm{SV},f} = \frac{|c_f^* \cap \mathrm{SV}|}{N_{\mathrm{SV}}}, \tag{14}$$

$$Q_{\mathrm{Syn},f} = \frac{\sum_{c \in \mathcal{C}_f : n_c \geq 2} |c \cap \mathrm{Syn}|}{N_{\mathrm{Syn}}}. \tag{15}$$

Thus $L_{\mathrm{SV},f}$ counts vesicles in the component selected by total agent count, and $Q_{\mathrm{Syn},f}$ counts synapsin in any component containing at least two agents, including Syn–Syn groups. The primary structural observable is the all-agent largest component fraction

$$L_{\mathrm{all},f} = \frac{|c_f^*|}{N_{\mathrm{SV}} + N_{\mathrm{Syn}}}, \tag{16}$$

while $Q_{\mathrm{Syn},f}$ and $L_{\mathrm{SV},f}$ describe local protein participation and collective vesicle participation, respectively.

Each outcome is averaged over the last $m = \max\{1, \lceil 0.30F \rceil\}$ of the F stored frames:

$$Y_{o,s} = \frac{1}{m} \sum_{f=F-m+1}^{F} O_{o,s,f}. \tag{17}$$

Here o indexes the observable, s the seed and f the saved frame; the force-condition index is suppressed. Each run is summarized before values are compared across seeds. For these trajectories, the 84-ms late window covers frames 421–600 (58.94–84 ms). The earlier comparison uses the same 30% rule on the first 200 frames, giving frames 141–200 (19.74–28 ms).

4.7 Force-factorial design

The factorial design fixes ionic strength at $I = 0.15\,\mathrm{mol/L}$ and nominal synapsin concentration at $30\,\mu\mathrm{mol/L}$. It includes all four combinations of the binary interaction switches, $E, H \in \{0, 1\}$. H1 uses the reference amplitudes $\epsilon_{\mathrm{SynSyn}} = 1.50$ and $\epsilon_{\mathrm{SynSV}} = 1.00$, corresponding to well depths of $1.50 k_{\mathrm{B}}T$ and $1.00 k_{\mathrm{B}}T$, respectively, with $\epsilon_{\mathrm{SVSV}} = 0$; H0 sets all hydrophobic amplitudes to zero. E1 uses the electrostatic kernel of eq. (6) and E0 sets that kernel to zero. In every condition, the

same van der Waals law and its parameters remain active.

Three seed blocks, 9310002–9310004, were completed in all four conditions. An earlier four-run pilot with seed 9310001 informed feasibility and the selection of structural observables. The 12 later trajectories provide the comparisons reported here. The 2-nm primary shell and the three sensitivity shells were selected before those runs. Fixed inputs are listed in table 4.

Table 4: Fixed inputs of the four-condition force comparison.

Quantity	Value	Model role
Integration step	0.0007 ms	Fixed in every condition
Duration	84 ms	120,000 physical steps
Saved-frame spacing	0.14 ms	600 saved frames per completed run
Temperature	310.15 K	Shared setting
Effective viscosity	0.35 Pa s	Literature-informed effective friction
Relative permittivity	74.2	Solvent approximation
Volume	0.10 µm^3	Closed spherical compartment
Synapsin concentration / count	30 μM / 1,807	Same in all conditions
Vesicle count	200	Same in all conditions
Vesicle core-radius input	20.45 nm	Bounds: 15–30 nm
Core-radius SD input	2.5 nm	Truncated heterogeneous input
Corona thickness	4.8 nm	Effective interpenetrable shell
Synapsin effective radius	3.972 nm	Coarse-grained input
SV / Syn surface potentials	-22.5 / $+16$ mV	Effective surface-potential inputs
Hamaker constant	5×10^{-21} J	Fixed van der Waals background
Surface regularization	0.3 nm	Numerical/force definition
Hydrophobic decay length	0.5 nm	Effective short-range input
H1 well depths Syn–Syn / Syn–SV	1.50 / 1.00 $k_\mathrm{B}T$	Syn–SV and Syn–Syn only; H0 sets zero
Core overlap / residual	0.5 / 0.005 nm	Distinct geometric quantities
Projection iteration cap	128	Early stopping enabled
Neighbor depth	Two cell layers	Candidate-pair force evaluation
Primary analysis shell	2 nm	Sensitivity: 0, 2, 4, 6 nm

E and H vary through the binary switches; the remaining inputs are held fixed.

4.8 Post hoc mobility analysis and experimental reference

The released source-data workbook for Hoffmann et al. reports synapsin motion in the presence of isolated vesicles and in living neurons [8]. Its Fig. 2h summarizes individual apparent diffusion coefficients evaluated over 2–4 ms using 1000-Hz tracking. We use the released with-SV mean as the external scale reference in the main figure.

For each simulated synapsin agent i, coordinates are converted from nm to μm and projected as $\boldsymbol{z}_{i,f} = (x_{i,f}, y_{i,f})$. With $F = 600$ and $\delta t_\mathrm{save} = 0.14$ ms, the time-averaged squared displacement

at lag k is

$$\widehat{\mathrm{MSD}}_i(k) = \frac{1}{F-k} \sum_{f=1}^{F-k} \|\boldsymbol{z}_{i,f+k} - \boldsymbol{z}_{i,f}\|^2. \tag{18}$$

We fit $\alpha_i + \beta_i \tau$ by ordinary least squares with a free intercept, using all 14 native lags between 2.10 and 3.92 ms and time in seconds. Then $\widehat{D}_i = \beta_i/4$ has units of $\mu\mathrm{m}^2/\mathrm{s}$. The run summary is the mean of $\widehat{D}_i$ over 1,807 synapsin agents. We then report the mean and SD across the three runs in each condition. The estimator is also applied to the final 180 frames to assess window sensitivity. S1 Text gives the agent-level distribution and fitted-slope diagnostics.

4.9 Post hoc species-resolved contact analysis

For every saved frame, the same 2-nm association graph is reconstructed from recorded centers and outer radii. Each unordered edge is counted once and classified as Syn–Syn, Syn–SV or SV–SV. Let $\mathcal{S}$ and $\mathcal{V}$ be the Syn and SV index sets. For Syn agent $i \in \mathcal{S}$, let

$$d_i^{\mathrm{SV}}(t) = \sum_{j \in \mathcal{V}} \mathbf{1}\{\|\boldsymbol{x}_i(t) - \boldsymbol{x}_j(t)\| \le a_i + a_j + 2\,\mathrm{nm}\}.$$

The fractions with at least one or at least two SV neighbors are $N_{\mathrm{Syn}}^{-1} \sum_{i \in \mathcal{S}} \mathbf{1}\{d_i^{\mathrm{SV}} \ge 1\}$ and $N_{\mathrm{Syn}}^{-1} \sum_{i \in \mathcal{S}} \mathbf{1}\{d_i^{\mathrm{SV}} \ge 2\}$. The reciprocal SV coverage counts SVs with at least one Syn neighbor, divided by N_{SV}. Counts and fractions are first averaged within each run over its last 180 frames, and only then summarized across seeds. The analysis uses the same saved trajectories and 2-nm shell as the structural comparison.

4.10 Seed-level statistical analysis

Each complete trajectory is one replicate. Observations of frames, agents and clusters are nested within it [31]. The three seed identifiers define blocks across conditions; their role is to pair run-level comparisons.

Each structural observable is first averaged over the final 180 frames of a trajectory. We report the mean and sample SD of the three run-level values for each condition. For a comparison of conditions A and B, the paired difference is $d_s = Y_{B,s} - Y_{A,s}$; its mean and sample SD summarize the three seed-wise differences. The E-by-H interaction is calculated within each seed block using

eq. (1). Individual seed values are shown in the figures and S1 Text. We use the size and consistency of paired differences to interpret the results.

4.11 Use of AI

The authors used ChatGPT (OpenAI) for language revision and assistance with code debugging, and Claude (Anthropic) for feedback on early drafts. The authors checked the code and manuscript and take responsibility for the final content.

Author contributions

Shokoofeh Akbari: Conceptualization, Methodology (mathematical formulation), Software, Investigation, Formal analysis, Data curation, Visualization, and Writing–original draft.
Oleg Shupliakov: Conceptualization (biological and LLPS input) and Writing–review and editing.
Yousef Jamali: Conceptualization, Methodology (scientific interpretation), Supervision, Funding acquisition, and Writing–review and editing.

Data and code availability

The numerical data underlying the figures and tables are provided in the Supporting Information. The reproducibility package contains the simulator and the 12 condition configurations, together with analysis scripts, processed per-seed results, numerical diagnostics and trajectory checksums. It is maintained at `https://github.com/BioMathLab-TMU/simulationsv-electrostatic-hydrophobic-response`, version `manuscript-84ms-20261030`. The original 28-ms inputs remain in release `v1.0.0`; the updated version adds the common 84-ms analysis inputs and reproduction scripts. Original author-generated code is licensed under MIT; author-generated data and tables are licensed under CC BY 4.0. Third-party code and experimental data retain their original licenses. The independent experimental source data of Hoffmann et al. are available at `https://doi.org/10.6084/m9.figshare.24179445.v1`.

Competing interests

The authors declare no competing interests.

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Supporting information captions

S1 Text. Supporting methods and numerical results. Per-seed structural endpoints, paired primary-interaction estimates, association-shell sensitivity, interface definitions and complete run-level interface outcomes, together with statistical and software validation details. (PDF)

S1 Data. Numerical data underlying the figures and tables. Per-seed structural observables, paired contrasts, graph-shell sensitivity, time-series inputs, interface diagnostics and mobility comparison values. (ZIP)